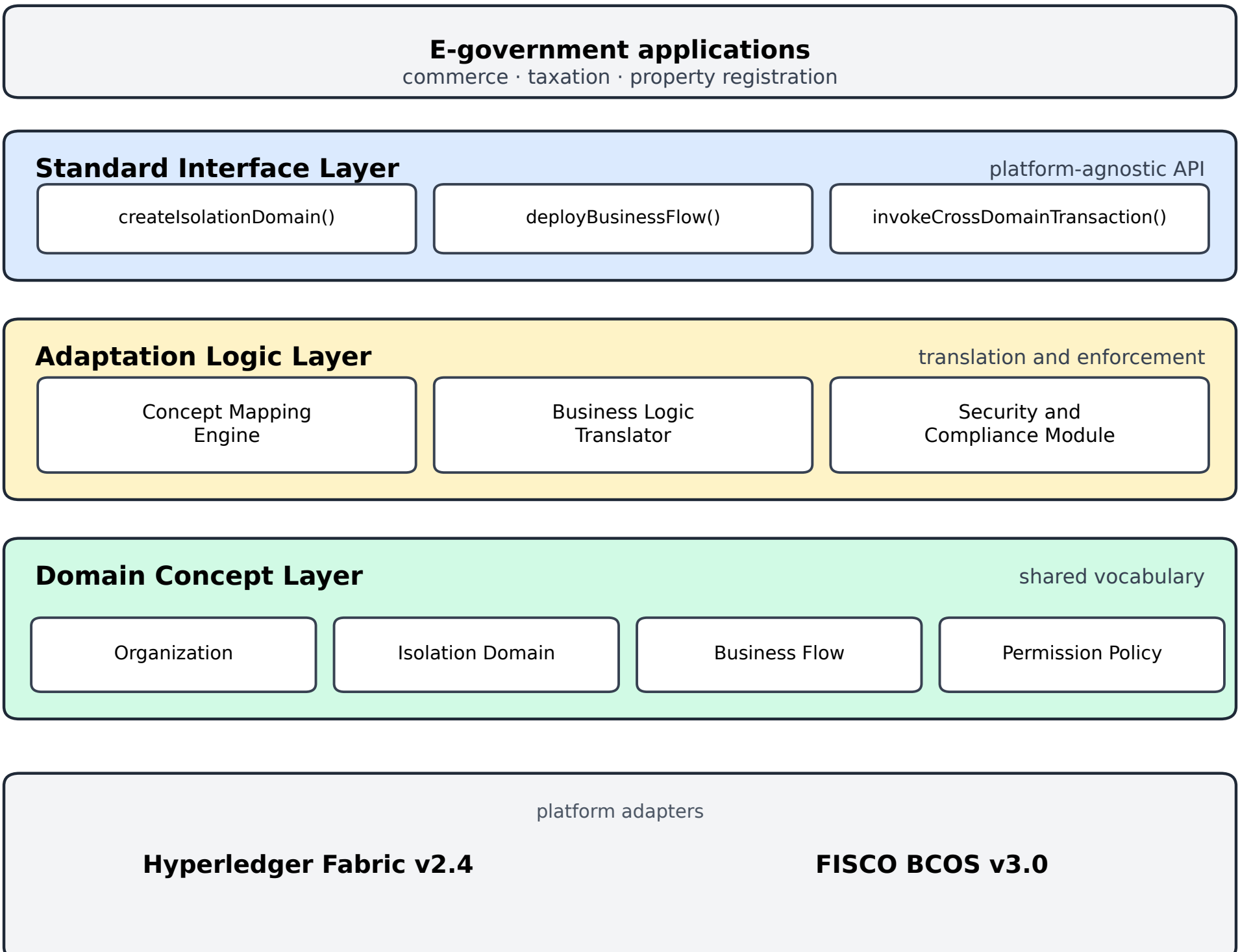Three-layer architecture of the DSAL prototype